Entropy and Distance of Random Graphs with Application to Structural Pattern Recognition

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Abstract—The notion of a random graph is formally defined. It deals with both the probabilistic and the structural aspects of relational data. By interpreting an ensemble of attributed graphs as the outcomes of a random graph, we can use its lower order distribution to characterize the ensemble. To reflect the variability of a random graph, Shannon's entropy measure is used. To synthesize an ensemble of attributed graphs into the distribution of a random graph (or a set of distributions), we propose a distance measure between random graphs based on the minimum change of entropy before and after their merging. When the ensemble contains more than one class of pattern graphs, the synthesis process yields distributions corresponding to various classes. This process corresponds to unsupervised learning in pattern classification. Using the maximum likelihood rule and the probability computed for the pattern graph, based on its matching with the random graph distributions of different classes, we can classify the pattern graph to a class.

Index Terms—Attributed graph, distance, entropy of random graph, graph, probability distribution, random graph, structural pattern recognition, synthesis of graphs, unsupervised classification.

I. Introduction

In structural pattern recognition, a pattern can be represented by a set of primitives and the relations among them; this can be a graph with the primitives as vertices and the relations among the primitives as arcs. Some proposed graph representations of patterns are the picture description language (PDL) and picture languages developed by Shaw [1]-[3], the primary graphs and region adjacency graphs (RAG) proposed by Pavlidis [4]-[6], and the attributed relational graphs (ARG) introduced by Tsai and Fu [7].

To compare attributed graphs, a similarity criterion or graph distance measure can be used during the graph matching process. Sanfeliu and Fu [8] group the conventional graph distance measures into the following two major categories.

1) Distances Using Feature Vector: By representing the principal features of a graph (such as number of vertices with various degrees, number of triangles, and so forth) as an n-dimensional vector, Euclidean distance can be adopted.

2) Distances Using Transformation Cost: The minimum number of modifications required to transform one graph into another can be used as their distance [8].

The first of the above measures generally satisfies the metric properties. However, its application is limited to graphs on which specific features can be defined easily. The second measure is intuitively simple and attractive. However, in general, it does not satisfy the metric properties and usually has exponential time complexity in its computation. In both definitions, information about the variability of graph attributes is not explicitly used.

When uncertainty exists in structural patterns either due to "noise" or minor deformation, a probabilistic description of the pattern ensemble is necessary. We therefore adopt the concept of random graph [9] for the classification and interpretation of graph or subgraph patterns.

II. Definitions of Attributed Graphs and Random Graphs

The basic concepts and definitions given below are adapted from [7] and [9] with modifications to comply with the commonly accepted notations [10].

A. Attributed Graphs

Attributed graphs have recently been introduced to render a more straightforward representation of structural patterns [7]. In this representation, a vertex with attributed values is used to represent a pattern primitive and an attributed arc is used to represent the relation between them.

Let $Z = \{z_i \mid i = 1, 2, \cdots, I\}$ be a nonempty and finite set of possible attributes for describing pattern primitives (vertex attributes) and, for each i, $S_i = \{s_{ij} \mid j = 1, 2, \cdots, J\}$ be the set of possible attribute values associated with z_i.[1] Let $L_v = \{(z_i, s_{ij}) \mid i = 1, \cdots, I; j = 1, \cdots, J_i\}$ be the set of legal attribute-value pairs. A pattern primitive π is just a specification of (unique) values for some of the attributes, i.e., π is a subset of L_v having at most one ordered pair for any particular z_i. Let Π denote the set of all possible pattern primitives.

Similarly, let $F = \{f_i \mid i = 1, 2, \cdots, I'\}$ be a nonempty finite set of possible relational attributes (edge attributes) between primitives, and T_i be the set of possible values of f_i. Let $L_a = \{(f_i, t_{ij}) \mid i = 1, \cdots, I'; j = 1, \cdots, T_i'\}$. A relation w is defined to be a subset of L_a which is a specification of (unique) values for some of the relational attributes. Let θ denote the set of all relations. The attribute value of a relation w is defined to be the n-tuple of values assigned by w.

Definition 1: An *attributed graph* G over $L = (L_v, L_a)$, with an underlying graph structure $H = (N, E)$, is defined to be a pair (V, A) where $V = (N, \mu)$ is called an *attributed vertex set* and $A = (E, \delta)$ is called an *attributed arc set.* The mappings μ: $N \to \Pi$ and δ: $E \to \theta$ are called *vertex interpreter* and *arc interpreter*, respectively.

We can consider G to consist of two parts: 1) $H = (N, E)$ which denotes its structure, and 2) $\Delta = (\mu, \delta)$ which describes

[1] These values may be nominal, ordinal, continuous, logical, etc.

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its attribute values. We call H the *structure* of G and Δ the *attribute* of G, respectively. For example, the relational structure in Fig. 1(b) depicts the structural aspect of the scene in Fig. 1(a), while Fig. 1(c) lists its attribute information.

To render a formal framework for comparison of attributed graphs, several basic definitions are introduced below.

Definition 2: The attribute matrix $M(G)$ of G with order n is an $n \times n$ matrix $[m_{ij}]$ such that

1) $m_{ij} = \mu(v_i)$, the vertex attribute value, if $i = j$;

2) $m_{ij} = \delta(a)$, the arc attribute value, if $i \neq j$ and $a = (v_i, v_j)$ is an arc of G; and

3) $m_{ij} = \phi$, otherwise.

Definition 3: Two attributed graphs $G_1 = (V_1, A_1)$ and $G_2 = (V_2, A_2)$ are said to be *structurally isomorphic* if there exists an isomorphism T: $H_1 \to H_2$ where $H_1 = (N_1, E_1)$ and $H_2 = (N_2, E_2)$ represent the structural aspects of G_1 and G_2, respectively. G_1 and G_2 are said to be *completely isomorphic*, written $G_1 = G_2$, if there exists an attribute value preserving structural isomorphism T between G_1 and G_2.

For example, the attributed graph in Fig. 2(b) representing the scene in Fig. 2(a) is structurally isomorphic to the graph in Fig. 1(b), but they are not completely isomorphic since the corresponding attribute values are not preserved. For matching graphs not of the same order, the null graph is introduced to extend the order of a graph.

Definition 4: A *null graph* is a graph $G_N = (V_N, A_N)$ where V_N, called a *null vertex set*, and A_N, a *null arc set*, are sets of elements Φ with null attribute values ϕ. A complete null graph of order k is denoted by G_{Nk}, and $m(G_{Nk})$ is a $k \times k$ matrix with all entries having value ϕ.

Definition 5: A graph $G = (V, A)$ of order n can be *extended* to form a complete graph $G' = (V', A')$ of order k, $k \geqslant n$, by adding vertices and arcs with null attribute values. We call G' the *k-extension* of G, and G the *n-reduction* of G'.

Fig. 3 illustrates an extension of a graph of order 4 to one of order 5. Here the dotted vertex and edges in G' denote a null vertex and null edges, respectively.

B. Random Graphs

The definition adopted here is from [9].

Definition 6: A *random graph* is a pair of tuples $R = (W, B)$ such that

1) W, representing the random vertex set, is an n-tuple $(\alpha_1, \alpha_2, \cdots, \alpha_n)$ where each α_i, called a random vertex, is a random variable;

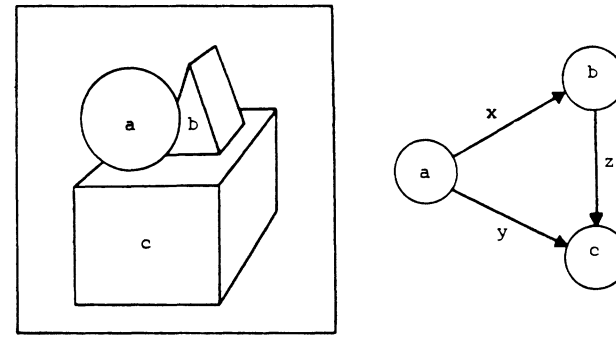
2) B, called the random arc family is an m-tuple $(\beta_1, \beta_2, \cdots, \beta_m)$ where each β_j, called a random arc, is also a random variable; and

3) associated with each possible outcome graph G of R and a graph monomorphism (or subgraph isomorphism) M: $G \to R$, there is a probability $p(G, M) = pr(R = M, Q = M)$, Q being the family of graph morphisms, which satisfies

$$p(G, M) \geqslant 0 \quad \text{for all } G \in \Gamma \tag{1}$$

$$\sum_{\Gamma} p(G, M) = 1 \tag{2}$$

where Γ is the range of R.

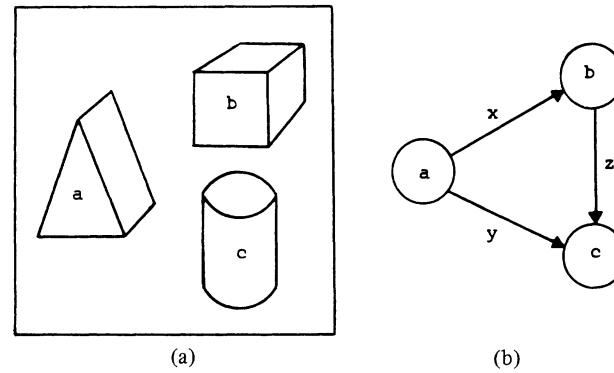


(a) (b)

vertex v	attribute value g(v)	arc a	attribute value h(a)
a	sphere	x	in front of
b	wedge	y	on top of
c	cube	z	on top of

(c)

Fig. 1. A relational structure and its representation by an attributed graph. (a) The scene. (b) Attributed graph. (c) Attribute values attached to (b).



(a) (b)

vertex v	attribute value g(v)	arc a	attribute value h(a)
a	wedge	x	to the left of
b	cube	y	to the left of
c	cylinder	z	above

(c)

Fig. 2. Another relational structure and its representation by an attributed graph. (a) The scene. (b) Attributed graph. (c) Attribute values attached to (b).

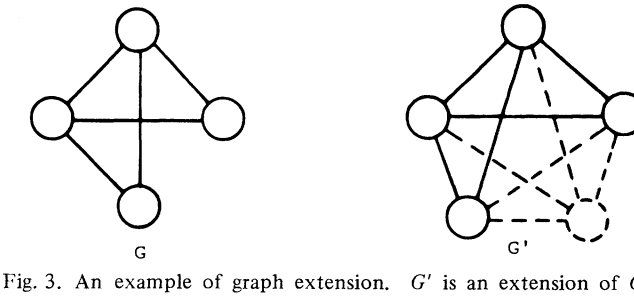


Fig. 3. An example of graph extension. G' is an extension of G to order 5.

We shall use the term *random element*, denoted by γ, to refer to either a random vertex or a random arc, or both, whenever a distinction between them is unnecessary. Hence, a random graph with n vertices and m arcs can also be expressed as an $(n + m)$-tuple with $n + m$ random elements, i.e.,

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$$R = (\gamma_1, \gamma_2, \cdots, \gamma_n, \gamma_{n+1}, \cdots, \gamma_{n+m}). \tag{3}$$

The range of each random element, which is referred to as an attribute, consists of a finite set of attribute values including the *null attribute value* ϕ that corresponds to a null outcome of γ. Therefore, if either endpoint of a random arc β has a null outcome, the outcome of the arc must also be null, i.e.,

$$\Pr\{\beta = \phi \mid \sigma(\beta) = \phi \quad \text{or} \quad \tau(\beta) = \phi\} = 1 \tag{4}$$

where $\sigma(\beta)$ and $\tau(\beta)$ are the initial and terminal random endpoints of β, respectively.

In the definition of attributed graph, if the z_i in Z are considered random variables taking values s_{ij} with probabilities p_{ij}, and similarly, the f_i in F are random variables taking values t_{ij} with probabilities q_{ij}, then the attributed graph just becomes a random graph in the obvious way.

With graph extension defined, we can slightly modify our representation of a random graph. Let $R = (W, B)$ be of order n. We can represent the *n-extension* of R by $R' = (W, B')$, such that, for each random arc $\beta' \in B'$, $\Pr\{\beta' = \phi\} = 1$, if $\beta' \notin B$. Furthermore, for each outcome graph $G = (V, A)$ of R with a monomorphism M: $G \to R$, we can consider its *n-extension* $G' = (V', A')$ as an outcome of R' with a monomorphism T: $G' \to R'$, such that, M is a restriction of T, and G' is structurally isomorphic to R'. It is obvious that the modification of R by R' does not affect either the probability of the outcome graph G or the monomorphism of G into R. That is,

$$\begin{aligned} p(G, M) &= \Pr\{R = G, Q = M\} = \Pr\{R' = G, Q = M\} \\ &= \Pr\{R' = G', Q = T\} = p(G', T). \end{aligned} \tag{5}$$

For simplicity of notations, we shall assume that all the random graphs are extended to complete random graphs and all outcome graphs are extended to the same order of their corresponding random graphs.

Using (4), we can partition the outcome set of a random graph into feasible and infeasible subsets. An infeasible outcome is one which is not a graph—an outcome consisting of a nonnull arc, one of whose endpoints is null. Such an outcome always has a probability 0. Hence, the feasible outcome subset constitutes a complete probability space. This verifies (2). In further discussions, the infeasible part is to be ignored in all outcomes of a random graph.

Definition 7: Two random graphs $R_1 = (W_1, B_1)$ and $R_2 = (W_2, B_2)$ are *completely isomorphic* iff there exists a structural isomorphism T: $R_1 \to R_2$, such that the probability distributions of their random vertices and random arcs are identical. R_1 and R_2 are said to be *equivalent*, denoted by $R_1 = R_2$, iff they are completely isomorphic.

Example 1: Fig. 4(a) represents a random graph R consisting of random vertices (A, B, C) and random arcs (X, Y, Z). The range of the random elements A, B, C, X, Y, and Z is $\{a, \phi\}$, $\{c, d\}$, $\{e, f\}$, $\{u, \phi\}$, $\{v, \phi\}$, and $\{x\}$, respectively, where ϕ denotes the null attribute value. Fig. 4(b) and (c) shows, among others, two feasible outcomes G_1 and G_2 of R with monomorphisms reflected by the vertex labels. Note that in G_2, the null arc $Y_2 = (A_2, C_2)$ is denoted by a dotted arrow with attribute value ϕ. Fig. 4(d), however, gives an infeasible outcome

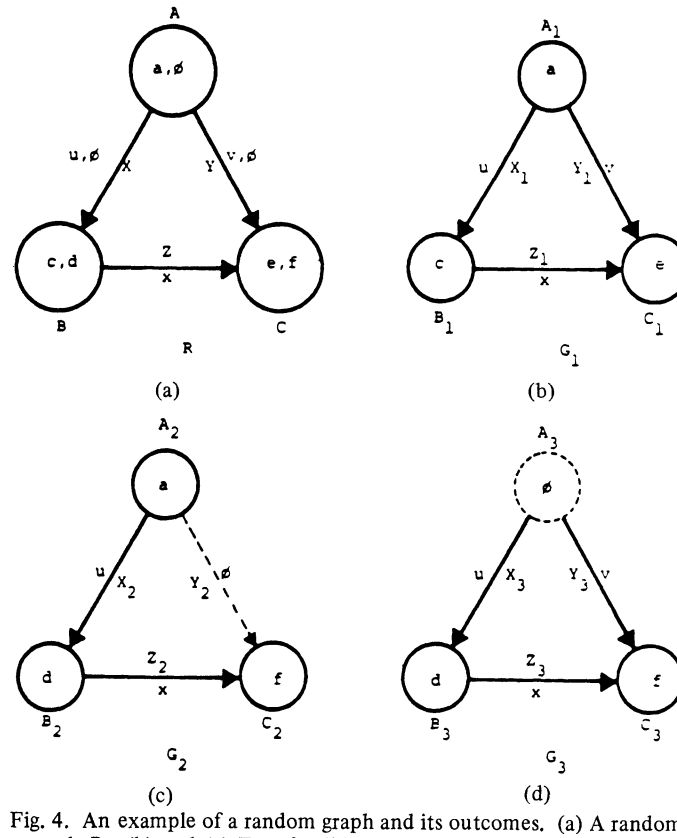

Fig. 4. An example of a random graph and its outcomes. (a) A random graph R. (b) and (c) Two feasible outcomes of R. (d) An infeasible outcome of R.

(G_3) of R where the dotted circle denotes the null (nonexisting) vertex A_3. Since A_3 does not exist in G_3, the two incident arcs X_3 and Y_3 do not have initial endpoints at all. Since an arc must have two endpoints in a graph, G_3 cannot be a graph, and thus its probability of being an outcome of R is 0.

III. Probability and Entropy of Random Graphs

Let T: $G \to R$ be the underlying monomorphism of an outcome G into R. Suppose that T is a restriction of S: $G' \to R$ where G' is an extension of G to the order of R. S can be denoted by two sets of mappings (M, N) where M: $V' \to W$ and N: $A' \to B$. $M(v')$ and $N(a')$ stand for the random vertex α and the random arc β. Thus, $M^{-1}(\alpha) = v'$, $N^{-1}(\beta) = a'$.

Definition 8: Let x and b (possibly null) be the outcomes of the random vertex α and the random arc β. Using (5), we define the probability of G with respect to R via T as

$$\begin{aligned} p(G, T) = p(G', S) = \Pr\{&(\alpha = x, \forall \alpha \in W), \\ &(\beta = b, \forall \beta \in B)\}. \end{aligned} \tag{6}$$

A. Estimation of Probability

In random graph applications, a lower order approximation of probability is required since the consideration of high-order joint probability space is impractical. However, if we assume that all random elements are mutually independent, we would preclude significant structural information from the graph. In this section, we present a probability estimation which takes into consideration some structural and contextual (attribute) information in the random graph.

We observe from (4) that certain degrees of dependency be-

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tween the arc and its endpoints exists. In order to express such dependencies explicitly, let us estimate (6) as

$$p(G, T) = \Pr\{\alpha = x, \forall\alpha \in W\} \Pr\{\beta = x, \forall\beta \in B \mid \alpha = x, \alpha \in W\}. \tag{7}$$

Next, we introduce the following assumptions:

1) the random vertices α are mutually independent,

2) a random arc β is independent of any random vertex other than its endpoints $\sigma(\beta)$ and $\tau(\beta)$, and

3) the conditional distributions $\{\beta \mid \alpha = x\}$ are mutually independent.

Based on the above assumption, (7) becomes

$$p(G, T) = \prod_{\alpha \in W} \Pr\{\alpha = x\} \times \prod_{\beta \in B} \Pr\{\beta = b \mid \sigma(\beta) = x, \tau(\beta) = y\}. \tag{8}$$

By putting $p(x) = \Pr\{\alpha = x\}$ and $p(b \mid x, y) = \Pr\{\beta = b \mid \sigma(\beta) = x, \tau(\beta) = y\}$ we write (8) as

$$p(G, T) = \prod_{\substack{\alpha \in W \\ M^{-1}(\alpha) = v' \\ \mu(v') = x}} p(x) \times \prod_{\substack{\beta \in B \\ N^{-1}(\beta) = a' \\ \delta(a') = b}} p(b \mid x, y) \tag{9}$$

where v' and a' are vertices and arcs in G'.

B. Entropy of Random Graph

In specifying the outcome variability, Shannon's entropy [11] is adopted.

Definition 9: The *entropy* $H(R)$ of a random graph $R = (W, B)$ is defined by

$$H(R) = -\sum_{(G, T)} p(G, T) \log p(G, T) \tag{10}$$

where the summation contains all outcome graphs G of R with the underlying monomorphism T. If $p(G, T) = 0$, the term $p(G, T) \log p(G, T)$ is defined as 0. The entropy $H(R)$ measures the uncertainty of R, or the variability of the outcome graphs of R. Using (9),

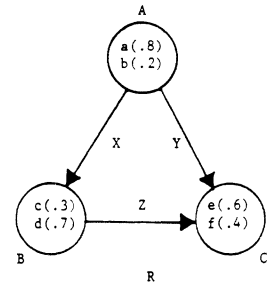
$$H(R) = \sum_{\alpha \in W} H(\alpha) + \sum_{\beta \in B} H(\beta) \tag{11}$$

where the summands are the entropies of the random vertices and random arcs, respectively. We can express

$$H(R) = -\sum_{x} p(x) \log p(x) - \sum_{x, y} p(x) p(y) \sum_{b} p(b \mid x, y) \log p(b \mid x, y). \tag{12}$$

Example 2: Fig. 5(a) is a problem graph with its vertex distributions. The conditional probability distributions of the random arcs are given in Fig. 5(b). Using (9), we obtain, as an illustration, the estimated probabilities for the outcomes G_1 and G_2 in Fig. 5(c) and (d) by

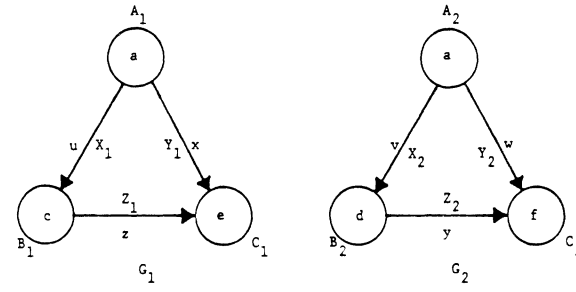
$$p(G_1, T_1) = [p(a)\, p(c)\, p(e)]\, [p(u \mid a, c)\, p(x \mid a, e)\, p(z \mid c, e)]$$



(a)

X	p	Y	p	Z	p
u|a,c	0.8	w|a,e	0.1	y|c,e	0
v|a,c	0.2	x|a,e	0.9	z|c,e	1.0
u|a,d	0.7	w|a,f	0.8	y|c,f	0.1
v|a,d	0.3	x|a,f	0.2	z|c,f	0.9
u|b,c	0.1	w|b,e	0.4	y|d,e	0
v|b,c	0.9	x|b,e	0.6	z|d,e	1.0
u|b,d	0.2	w|b,f	0	y|d,f	0.4
v|b,d	0.8	x|b,f	1.0	z|d,f	0.6

(b)



(c) (d)

Fig. 5. A random graph and two of its outcomes. (a) The random graph R. (b) The probability distribution of random arcs in R. (c) and (d) two outcomes G_1 and G_2.

$$= 0.104$$

$$p(G_2, T_2) = [p(a)\, p(d)\, p(f)] \cdot [p(v \mid a, d)\, p(w \mid a, f)\, p(y \mid d, f)] = 0.022.$$

Using (11) and (12), we obtain

$$H(R) = [H(A) + H(B) + H(C)] + [H(X) + H(Y) + H(Z)] = 4.271.$$

IV. Synthesis of Random Graphs with Known Morphism

Consider a class of attributed graphs $C = \{G_1, G_2, \cdots, G_m\}$. Suppose that associated with each $G_i = (V_i, A_i)$ in C there is a vertex mapping T_i: $V_i \to L$ which assigns each vertex $v \in V_i$ a distinct label from the set L.[2] We say that two graphs G_i and G_j are *equivalent under* L if $T_i \circ (T_j)^{-1}$ is a complete isomorphism between G_i and G_j. Therefore, we can partition C into equivalence classes $\{C_1, C_2, \cdots, C_n\}$ under L where $n \leqslant m$. Thus, each equivalence class C_i can be represented by a graph $G_i \in C_i$. Hence, we can represent C by a set of distinct graphs $D = \{G_1, G_2, \cdots, G_n\}$. For each G_i in D, we associate a frequency f_i indicating the size of C_i.

In this section, we first propose a method of synthesizing a set of attributed graphs into a random graph. Next, we generalize the method to synthesize a set of random graphs.

A. Synthesis of Attributed Graphs

Given $D = \{G_1, G_2, \cdots, G_n\}$ we can derive a random graph R satisfying the following conditions.

1) There exists a vertex labeling S: $R \to L$.

[2]Note that L is a set of distinct vertex labels, not vertex attribute values.

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2) There is a monomorphism M_i: $G_i \to R$ such that $T_i = M_i \circ S$.

3) The probability of each G_i can be estimated as the relative frequency of G_i in D, i.e.,

$$p(G_i, M_i) = \Pr\{R = G_i, Q = M_i\} = f_i \Big/ \sum_{i=1}^{n} f_i. \tag{13}$$

Let the vertex set corresponding with label k among G_i's in D be $U_k = \{v_{k1}, v_{k2}, \cdots, v_{kn}\}$ where $v_{ki} \in V_i$. Let α_k be the random vertex and let x be an arbitrary outcome associated with α_k. Then, $p_k(x)$ can be estimated by

$$p_k(x) = \Pr\{\alpha_k = x\} = \sum_{\substack{i=1 \\ \mu(v_{ki}) = x}}^{n} f_i \Big/ \sum_{i=1}^{n} f_i. \tag{14}$$

Similarly, we denote the set of the corresponding arcs among G_i's in D by $S_j = \{a_{j1}, a_{j2}, \cdots, a_{jn}\}$ such that, for $i = 1, 2, \cdots, n$, both the initial and terminal endpoints of a_{ji} have, respectively, identical labels. Let β_j be the random arc with outcomes being the values of a_{ji} in S_j. Then, the conditional probability that β_j assumes a value b given the endpoints values x and y can be estimated by

$$\begin{aligned} p_j(b|x,y) &= \Pr\{\beta_j = b \,|\, \sigma(\beta_j) = x, \tau(\beta_j) = y\} \\ &= \sum_{\substack{i=1 \\ \mu(\sigma(a_{ji})) = x \\ \mu(\tau(a_{ji})) = y \\ \delta(a_{ji}) = b}}^{n} f_i \Bigg/ \sum_{\substack{i=1 \\ \mu(\sigma(a_{ji})) = x \\ \mu(\tau(a_{ji})) = y}}^{n} f_i. \end{aligned} \tag{15}$$

Therefore, a random graph R consisting of the random vertices and random arcs with probability distributions defined in the form of (14) and (15) is more general than the one given in (13). We call a random graph R obtained from D using (14) and (15) a *synthesis* of D.

B. Synthesis of Random Graphs

Suppose that a set of random graphs $F = \{R_1, R_2, \cdots, R_m\}$ can be independently obtained from subgroups of a class of pattern graphs. Let n_i be the number of graphs in the ith subgroup. Suppose further that a set of morphisms $\{T_i | T_i: R_i \to L\}$ exists which maps each R_i into a common label set L. Then, we can obtain for F a synthesis which is a "super" random graph to encompass all R_i's in F.

Assume that $\{\alpha_{k1}, \alpha_{k2}, \cdots, \alpha_{km}\}$ and $\{\beta_{j1}, \beta_{j2}, \cdots, \beta_{jm}\}$ are the sets of the kth random vertices and jth random arcs among $\{R_i\}$ according to $\{T_i\}$. Let $\{p_{ki}(x)\}$ and $\{p_{ji}(b|x,y)\}$ denote the probability distributions of α_{ki} and β_{ji}, respectively. We can define the synthesis of the set of random vertices and random arcs as follows.

Definition 11: The *synthesis* α_k of a set of random vertices $\{\alpha_{k1}, \alpha_{k2}, \cdots, \alpha_{km}\}$ is defined to be a random vertex having the probability distribution

$$p_k(x) = \Pr\left\{\alpha_k = x \,\middle|\, \bigcup_{i=1}^{m} \alpha_{ki}\right\} = \sum_{i=1}^{m} q_i p_{ki}(x) \tag{16}$$

where

$$q_i = n_i \sum_{i=1}^{m} n_i \tag{17}$$

is the relative frequency (or *a priori* probability) of R_i in the combined sample after the merging of all R_i's, and $p_{ki}(x) = \Pr\{\alpha_{ki} = x\}$ is the probability of the random vertex α_{ki} having outcome x.

Definition 12: The synthesis β_j of a set of random arcs $\{\beta_{j1}, \beta_{j2}, \cdots, \beta_{jm}\}$ is defined to be a random arc having the following conditional probability distribution $\{p_j(b|x,y)\}$:

$$\begin{aligned} p_j(b|x,y) &= \Pr\left\{\beta_j = b \,\middle|\, \bigcup_{\substack{i=1 \\ \sigma(\beta_{ji}) = x \\ \tau(\beta_{ji}) = y}}^{m} \beta_{ji}\right\} \\ &= \left[\sum_{i=1}^{m} q_i r_{ji} p_{ij}(b|x,y)\right] \Big/ [p_k(x) p_l(y)] \end{aligned} \tag{18}$$

where

1) $r_{ji} = \Pr\{\sigma(\beta_{ji}) = x, \tau(\beta_{ji}) = y\} = p_{ki}(x) p_{li}(y)$,

2) $p_{ji}(b|x,y) = \Pr\{\beta_{ji} = b \,|\, \sigma(\beta_{ji}) = x, \tau(\beta_{ji}) = y\}$, and

3) $p_k(x) p_l(y) = \Pr\{\sigma(\beta_j) = x\} \times \Pr\{\tau(\beta_j) = y\} = \Pr\{\sigma(\beta_j) = x, \tau(\beta_j) = y\}$ [the probability of the synthesis random arc β_j with endpoints x and y obtained using (16)].

The random graph R composed of random vertices and random arcs defined above is called a *synthesis* of F according to the given morphism. Two random graphs R_1 and R_2, as shown in Fig. 6, are synthesized under a given morphism T. Here, R_1 and R_2 are supposed to have equal *a priori* probabilities $q_1 = q_2$. Fig. 7 shows the result of the synthesis.

So far, morphism between graphs is assumed known in the synthesis. When morphism is not known *a priori*, an optimal isomorphism between random graphs will be used.

V. Distance Measure Between Random Graphs

The scheme of combining independent random graphs is similar to the method of considering independent experiments as components of a single composite experiment, in which additional information obtained in later experiments is added to that of the previous ones [14]. An interesting property of combining probability distributions is that the entropy of the combined is never less than the weighted average of the entropies of the probability distributions. In this section, we will first illustrate such property in random graph synthesis and then use it as a criterion for defining optimal isomorphism between random graphs.

A. Increment of Entropy in the Synthesis of Random Graphs

Let $R = +(R_1, R_2, T)$ denote that $R = (W, B)$ is the synthesis of the random graphs R_1 and R_2 under a given morphism T, and $\gamma = +(\gamma_1, \gamma_2, T)$ denote that γ is the synthesis of the corresponding random elements γ_1 and γ_2 in R_1 and R_2, respectively, according to T. Then we have the following definition.

Definition 13: Let q_1 and q_2 be the relative frequencies in the samples associated with R_1 and R_2, respectively. The *increment of entropy* in γ is defined as follows:

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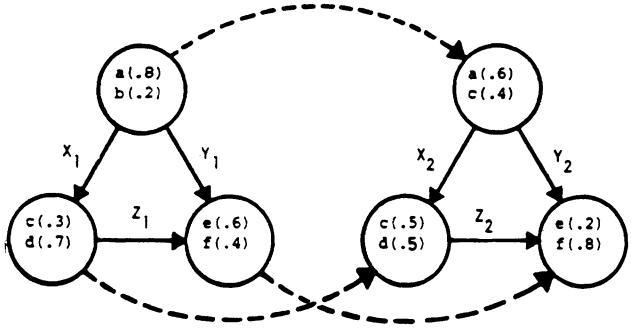


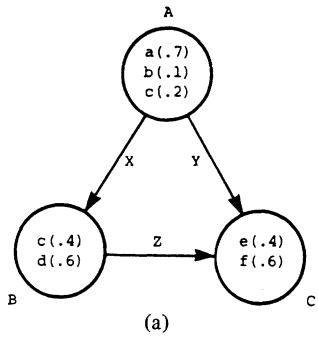
(a)

X_1	p	Y_1	p	Z_1	p
x|a,c	1.0	y|a,e	1.0	z|c,e	1.0
x|a,d	1.0	y|a,f	1.0	z|c,f	1.0
x|b,c	1.0	y|b,e	1.0	z|d,e	1.0
x|b,d	1.0	y|b,f	1.0	z|d,f	1.0

X_2	p	Y_2	p	Z_2	p
x|a,c	1.0	y|a,e	1.0	z|c,e	1.0
x|a,d	1.0	y|a,f	1.0	z|c,f	1.0
y|c,c	1.0	x|c,e	1.0	z|d,e	1.0
y|c,d	1.0	x|c,f	1.0	z|d,f	1.0

(b)

Fig. 6. Example of random graph synthesis. (a) T: $R_1 \to R_2$. (b) The probability distributions of random arcs in R_1 and R_2.



(a)

X	p	Y	p	Z	p
x|a,c	1.0	y|a,e	1.0	z|c,e	1.0
x|a,d	1.0	y|a,f	1.0	z|c,f	1.0
x|b,c	0.5	y|b,e	0.5	z|d,e	1.0
x|b,d	0.5	y|b,f	0.5	z|d,f	1.0
y|c,c	0.5	x|c,e	0.5		
y|c,d	0.5	x|c,f	0.5		

(b)

Fig. 7. The synthesis of the graphs in Fig. 6. (a) $R = +(R_1, R_2, T)$. (b) The probability distribution of random arcs in R.

$$H'(\gamma) = H(\gamma) - [q_1 H(\gamma_1) + q_2 H(\gamma_2)]. \tag{19}$$

Analogously, the *increment of entropy* in R is defined as

$$H'(R) = H(R) - [q_1 H(R_1) + q_2 H(R_2)]. \tag{20}$$

Note that (20) is equivalent to the sum of the increment of entropy in all random elements γ in R, i.e.,

$$H'(R) = \sum_{\gamma \in R} H'(\gamma), \quad \text{or} \tag{21}$$

$$H'(R) = \sum_{\alpha \in W} H'(\alpha) + \sum_{\beta \in B} H'(\beta) \tag{22}$$

where W and B are the random vertex set and random arc set of R, respectively.

Property 1: The increment of entropy in $\gamma = +(\gamma_1, \gamma_2, T)$ is bounded

$$0 \leqslant H'(\gamma) \leqslant 1. \tag{23}$$

The lower bound can be proved by observing that $-p \log p$, $0 \leqslant p \leqslant 1$, is a concave function. The upper bound is also obvious from the properties of entropy. Through a computer simulation, we also found that $H'(\gamma)$ is no greater than 1. We conjecture this may be a tighter upper bound. Property 1 immediately leads to another property, as follows.

Property 2: If R is of order N, then

$$0 \leqslant H'(R) \leqslant N. \tag{24}$$

The increment of entropy is the synthesis of two random elements (belonging, respectively, to R_1 and R_2) can be used to depict the increase of uncertainty when they are merged. Where they have the same *a priori* probability and when each of them assumes only a single but different value, the synthesis will have an uncertainty with value of log 2 (= 1) and this may be the maximum possible increase of uncertainty as conjectured. If two random elements have exactly the same probability distribution before merging, there will be no increase in uncertainty since the distribution after the synthesis remains unchanged.

Example 3: Fig. 8(a) illustrates the synthesis of two different "deterministic" random elements having the same relative frequency. The increment of entropy in the synthesis obviously is 1. Fig. 8(b) shows the synthesis of two random elements having the same probability distribution. Note that if the original probability distributions have the same entropy value, the *a priori* probabilities q_1 and q_2 would not affect the result.

B. Weighted Increment of Entropy

To take into account the difference in likelihoods of variations between outcomes of γ_1 and γ_2 we introduce the following definitions.

Definition 14: Let $s = (s_1, s_2, \cdots, s_n)$ and $t = (t_1, t_2, \cdots, t_2)$ be outcomes of two random elements γ_1 and γ_2, respectively. We define the *difference* $d(s_i, t_i)$ between the corresponding attribute values s_i and t_i associated with attribute c_i in s and t as follows:

$$d(s_i, t_i) = |s_i - t_i| / r_i \tag{25}$$

where r_i is the value range of attribute i. For attributes with nominal (or nonnumerical) values, we can either use the Hamming distance or further define $d(s_i, t_i)$ as follows:

$$d(s_i, t_i) = 1 - \tfrac{1}{2}(\Pr\{s_i \to t_i\} + \Pr\{t_i \to s_i\}) \tag{26}$$

where $\Pr\{x \to y\}$ denotes the probability of the nominal value x being deformed into the nominal value y.

Definition 15: The *variation weight* w_t between the outcome of γ_1 and that of γ_2 is defined to be

$$w_t = \sum_s \sum_t \Pr\{\gamma_1 = s\} \Pr\{\gamma_2 = t\} \, d(s, t) \tag{27}$$

where

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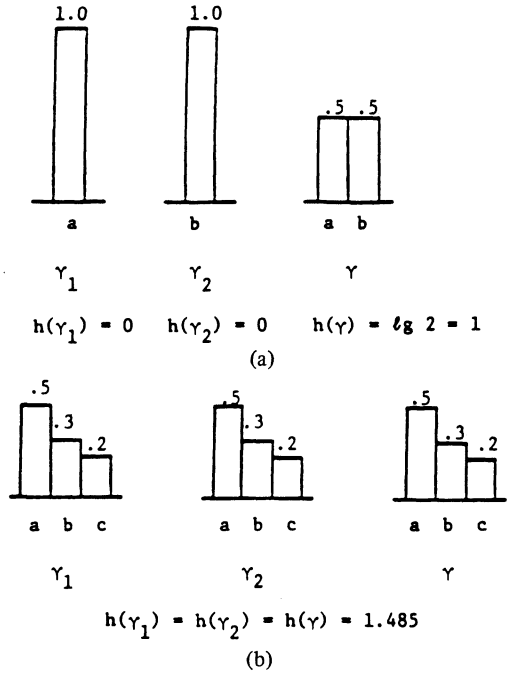

Fig. 8. Examples of the syntheses of graph elements. (a) The synthesis of two different "deterministic" random elements. (b) The synthesis of two random elements having the same distribution.

$$d(s, t) = \sum_i w_i \, d(s_i, t_i) \tag{28}$$

is the weighted difference between s and t, with w_i being the weight corresponding to attribute i. $d(s_i, t_i)$ can be obtained by either (25) or (26), depending on the type of values associated with attribute i.

Note that $w_t \geqslant 0$, and has a value of 0, iff there exists outcome e such that $p_1(e) = p_2(e) = 1$. In other words, when both elements are "deterministic" and have the same outcome space, they are identical, and hence, we should impose no variation weight. Furthermore, from (25) and (26), it is obvious that $d(s, t)$ in (28) is in $[0, 1]$. Therefore, w_t is also bounded in the range of $[0, 1]$.

Definition 16: The *weighted increment of entropy* $H''(\gamma)$ in random element γ is defined as follows:

$$H''(\gamma) = w_t H'(\gamma) \tag{29}$$

where w_t is the variation weight of γ as in (27) and $H'(\gamma)$ is the (unweighted) increment of entropy in γ (19).

Using (22) and (29), we can readily derive the *weighted increment of entropy* $H''(R)$ in the synthesis $R = (W, B)$ of two random graphs R_1 and R_2 as

$$H''(R) = \sum_{\alpha \in W} H''(\alpha) + \sum_{\beta \in B} H''(\beta). \tag{30}$$

Property 3: The weighted increment of entropy in the synthesis γ of random elements is also bounded

$$0 \leqslant H(\gamma) \leqslant 1. \tag{31}$$

Property 4: The weighted increment of entropy in the synthesis R of random graphs of order N is bounded

$$0 \leqslant H(R) \leqslant N^2. \tag{32}$$

C. Distance and Optimal Isomorphism

From the systems point of view, if we consider R_1 and R_2 as independent parts of a system, and their synthesis $R = +(R_1, R_2, T)$ as the system as a whole, then the increment of entropy signifying the difference between the average entropy of R_1 and R_2 and the entropy of R can be adopted as a measure of organization (structure) in the system (pattern) [13]. Hence, the entropy increment in the synthesis should be low if the independent parts pertain to the same class. In other words, we can use $H(R)$ as a criterion to discriminate the goodness of matching between R_1 and R_2 under the morphism T that yields R.

Definition 17: The *distance* between two random graphs R_1 and R_2 is defined to be

$$d(R_1, R_2) = \min_T \{H''(R) \mid R = +(R_1, R_2, T)\} \tag{33}$$

where the minimization takes over all the possible structural isomorphisms T. One of such T denoted by T^* which yields a minimum $H''(R)$ is called an *optimal isomorphism* between R_1 and R_2. Obviously, T^* may not be unique.

Property 5: Let the order of the synthesis of R_1 and R_2 be N. The distance measure $d(R_1, R_2)$ has the following properties:

1) $0 \leqslant d(R_1, R_2) \leqslant N^2$;
2) $d(R_1, R_2) = 0$ iff $R_1 = R_2$ (complete isomorphism); and
3) $d(R_1, R_2) = d(R_2, R_1)$.

Unfortunately, this distance measure generally does not satisfy triangular inequality of a metric due to the averaging effect weighted by the *a priori* probabilities.

In order to compare random graphs of different orders, it is desirable to normalize the absolute distance, making it a measure of relative similarity. Since the weighted increment of entropy in a synthesis of two random graph is bounded by 0 and N^2 (32), we arrive at the following definition.

Definition 18: Let $d(R_1, R_2)$ be the distance defined by an optimal isomorphism between R_1 and R_2 of order N. We define the *similarity* $s(R_1, R_2)$ between R_1 and R_2 to be

$$s(R_1, R_2) = 1 - d(R_1, R_2)/N^2. \tag{34}$$

Property 6: The similarity measure defined above has the following properties:

1) $0 \leqslant s(R_1, R_2) \leqslant 1$;
2) $s(R_1, R_2) = 1$ iff $R_1 = R_2$ (complete isomorphism); and
3) $s(R_1, R_2) = s(R_2, R_1)$.

Since an attributed graph can be treated as a special case of a random graph, the above distance (or similarity) can be defined for 1) two attributed graphs, 2) an attributed graph and the probability distribution of a random graph, and 3) the probability distributions of two random graphs. These measures have been applied to structural pattern recognition (Section VII).

VI. Synthesis of Random Graphs with Morphism Unknown

Since $H(R)$ is a measure of the variability of its outcomes, a low entropy value indicates greater similarity among its outcome graphs. Furthermore, $H(R)$ depends on the interpreta-

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tion of each graph in the ensemble as outcomes of R, i.e., on the choice of the morphism T among them. Assuming that graphs given in an ensemble retain considerable similarity, we can use entropy as an objective function to direct the labeling and estimation process. Hence, it is possible to obtain a minimum entropy in synthesizing R by solving the optimization problem [9]

$$\min_T H(R). \tag{35}$$

However, when the ensemble size is large and the order of the graphs is high, we need a hierarchical method for approximating the optimization process. At each stage, a local optimum is obtained. And eventually, a final result is selected among the local optima. The synthesis obtained is then adopted as the approximation of the random graph which represents the class pattern of the entire ensemble.

Aside from simplicity in computation, the hierarchical approximation has another advantage. It is capable of detecting subclasses in the ensemble. Without such capability, the synthesis process may lead to unnatural and misleading results.

A. A Hierarchical Synthesis Process

Let $D = \{G_1, G_2, \cdots, G_n\}$ be the observed graphs in an ensemble. Without loss of generality, each G_i in D can be treated as a random graph R_i. We can thus equate D to a set $F = \{R_1, R_2, \cdots, R_n\}$, where each R_i in F is a random graph representation of the attributed graph G_i in D. Let R_{ij} denote the synthesis of R_i and R_j under the optimal isomorphism T_{ij}. Then, the distance d_{ij} between them is $H''(R_{ij})$. We use two $n \times n$ matrices[3] $L = [T_{ij}]$ and $M = [d_{ij}]$, to direct the synthesis of F. Obviously, M is a symmetric matrix with diagonal elements $d_{ii} = 0$ for all i.

In the iterative process, those pairs of distinct graphs in F with a minimum distance in M are synthesized and replaced by their synthesis. After reindexing, F is reduced to $F = \{R_1, R_2, \cdots, R_m\}$, $m \leqslant n$. This procedure is repeated until one random graph, the synthesis of the set, remains, or until it is terminated by a thresholding criterion. The final remaining random graph(s) represents the class pattern (or subclass patterns), encompassing all possible variations among its members in D.

B. Thresholding

When a class of patterns is composed of several subclasses, we can introduce a threshold to determine the splitting condition. If the threshold criterion between two random graphs exceeds the threshold, instead of synthesizing them, we allow them to coexist as tentative representations of separate subclass patterns of the same class.

A readily usable criterion for thresholding is the similarity measure s_{ij} between graphs R_i and R_j. It is a normalized measure ranging from 0 to 1. An appropriate point t in this range can be prescribed so that only random graph pairs with $s_{ij} \geqslant t$ will be synthesized.

At the beginning of the process, it is difficult to prescribe the value t. To overcome this, a dynamic thresholding criterion is proposed. Let S_i and S_j be lists of all previous similarity measures that lead to the synthesis of R_i and R_j, respectively. We define the *average similarity measure* of R_i and R_j to be

$$s^* = \left(\sum_{s \in S_i} s + \sum_{s \in S_j} s\right) \Big/ (|S_i| + |S_j|). \tag{36}$$

Now, the dynamic threshold t^* can be defined to be

$$t^* = r \times s^* \tag{37}$$

where r again is a prescribed rate with a value in $[0, 1]$.

In applications, R_i and R_j will be synthesized only if $s_{ij} \geqslant t^*$. Apparently, t^* becomes tighter when the set of graphs sustains higher similarity. This variability characteristic of t^* makes it flexible and useful when no *a priori* knowledge of the class of graphs is available, although the determination of r in (37) depends on the informed knowledge of the problem domain.

C. The Graph Synthesis Algorithm

In this section, we present the hierarchical graph synthesis algorithm (HGS) in pseudocode.

Input:
1) A class of graphs $D = \{G_1, G_2, \cdots, G_n\}$;
2) a) A prescribed number of classes, n_k; or
   b) i) A prescribed graph similarity threshold t; and/or
      ii) A prescribed rate for average similarity r.

Output:
1) A reduced set of random graphs, $F = \{R_1, R_2, \cdots, R_m\}$, $1 \leqslant m \leqslant n$;
2) The labelings of the outcome graphs in D.

Steps:

```
begin {HGS}
  begin {initiation}
  . F := D;
  . define all R_i in F to be "new";
  . current_number := number of entries in D;
  . for i := 1 to n do S_i := empty;
  . L := n × n empty matrix;
  . M := n × n empty matrix;
  end; {initiation}
  if current number ≥ 2 then repeat
  . for each pair of graphs R_i and R_j in F do
  . if R_i = "new" or R_j = "new" then begin
  . . find optimal isomorphism T_ij and save T_ij in L;
  . . compute distance d_ij and save d_ij in M;
  . . define R_i and R_j to be "old";
  . end; {for-if}
  . min_diff := minimum distance value in M;
  . for each d_ij in M do begin
  . . if d_ij ≤ min_diff and R_i, R_j not synthesized then
  . . begin
  . . . compute s_ij;
  . . . if S_i = empty and S_j = empty then s* := s_ij
  . . . else begin
```

[3] The term "matrix" here is used in a general sense. For example, instead of scalar values, the entries may contain pointers to lists.

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```
. . . . . compute s* from S_i and S_j; {see (36)}
. . . . . t* := r × s*;
. . . end; {else}
. . . if s_ij ≥ t and s_ij ≥ t* then begin
. . . . . R_ij := + (R_i, R_j, T_ij); {synthesize R_i, R_j}
. . . . . F := F - R_i; {deletion}
. . . . . label G_k, the original graph in D
. . . . . . corresponding to R_i, according to T_ij;
. . . . . R_j := R_ij;
. . . . . reset R_j to be "new";
. . . . . S_j := S_i, S_j, s_ij; {concatenation}
. . . end; {if s_ij}
. . end; {if d_ij}
. end; {for}
. re_index all remaining graphs {R_1, R_2, ···, R_m} in F;
. re_index elements in M corresponding to F;
. previous_number := current_number;
. current_number := number of graphs remaining in F;
until current_number = previous_number, or n_k, or l;
end. {HGS}
```

The time complexity of the above algorithm is of order $O(m^3)$ times the order of the average time complexity of the optimal graph isomorphism.

VII. Applications of Random Graph in Pattern Recognition

In this section, we use a simple example to illustrate the applicability of the random graph in structural pattern recognition.

A. Using Random Graphs to Characterize Pattern Classes

Generally, a random graph R, which is the synthesis of a cluster of individual attributed graphs, reflects the class probabilistic configuration of the cluster. If the graphs are designated as belonging to the same pattern class, the synthesis process is referred to as *supervised*. If the class memberships for a given set of graphs are not known, the synthesis which generates more than one random graph is considered an *unsupervised* classification.

The distribution of a random graph is referred to as the *structural probability distribution* of a pattern class. One of the advantages of using such a distribution is that it can also be used as a base graph in graph matching for classification purposes [19]. Other major advantages are as follows.

1) The random graph distribution encompasses both the structural and the statistical information of the entire class of graphs. It is a better representation of the class than, say, the cluster prototypes.

2) The random graph allows varied or deformed patterns as members of the class although they were not included in the training set.

3) The classification of a pattern can be achieved by comparing the pattern to each random graph in a class.

B. Classification Procedure

Suppose that $R_1, R_2, \cdots, R_n$ represent n different classes (or subclasses). Let G be an attributed graph representing an unknown pattern. To classify G, we can use either the nearest neighbor rule or the maximum likelihood rule. For either rule, we first find a graph optimal isomorphism T_i: $G \to R_i$, $i = 1, \cdots, n$. Depending on the criterion used in finding T_i, the distances $d(G, R_i)$ between G and R_i, or the probabilities $p(G, T_i)$ of G being an outcome of R_i, can be obtained. If for some R_k, T_k does not exist (e.g., the matching cost (distance) is infinity or the probability is 0), the class k can be excluded. Using the nearest neighbor rule, we classify G to class i, iff

$$d(G, R_i) < d(G, R_j), i \neq j, \quad \text{for all } j. \tag{38}$$

Using the maximum likelihood rule, we classify G to class i iff

$$p(R_i)\, p(G, T_i) > p(R_j)\, p(G, T_j), i \neq j, \quad \text{for all } j \tag{39}$$

where $p(R_j)$ is the *a priori* probability of class j characterized by R_j, and $p(G, T_j)$, based on (11), is the estimated probability of G being an outcome of R_j.

C. An Illustrative Example in Character Recognition

In this section, we use English character recognition to illustrate the use of random graphs in structural pattern recognition. A pseudorandom number generator is used to simulate stroke orientation variation in the handwritten English letters F, H, and E. Twenty characters are then generated for each letter. Among the 60 characters, all 47 distinct variations are taken and listed in Fig. 9, where the numbers to the left and right of ":" denote the character index and its frequency of occurrence, respectively. The attribute codes for character primitives and primitive relations are defined in Fig. 10. Fig. 11 shows the graphs representing the characters indexed 1, 21, and 31.

The graph synthesis algorithm is adopted for clustering. First, pairwise distances among the attributed graphs are calculated and tabulated. The pair(s) of graphs with the shortest distance are then merged. Next, new distances between the merged graphs and the remaining graphs are computed. The closest pair(s) are merged again. The process continues as shown in the hierarchical tree [Fig. 12(a)], from which we observe that all 47 distinct characters are successfully clustered into their respective letter classes.

Using the three known distributions representing variations of letters F, H, and E in Fig. 12(b), we can classify the attributed graph of an unknown sample to one of the pattern classes. For the three unknown patterns (Fig. 13), not contained in the ensemble for the clustering process and not used in the training process, we classify each of them into its correct class (Fig. 14) using both rules.

VIII. Conclusion

In this paper, we have formally defined the attributed graph and the random graph. We have also discussed the probability distribution estimation and the entropy of random graphs. Taking the "deterministic" attributed graph as a special case of the random graph, we have proposed a distance between random graphs. With this, we are able to apply unsupervised classification to an ensemble of patterns represented by attributed graphs. When random graphs of known classes are given,

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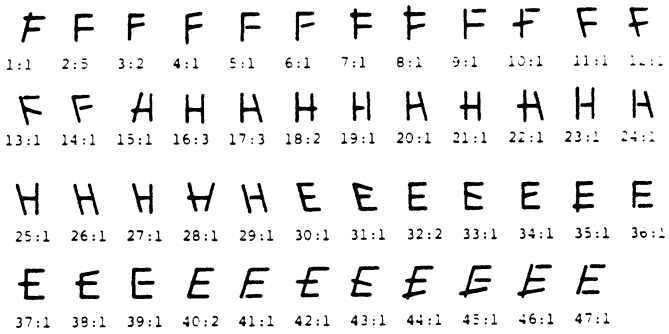
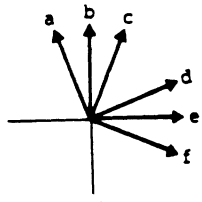


Fig. 9. Variations of handwritten English letters F, H, and E.



(a)

Code	Description	Diagram
u	above	
v	bending	or
w	branching	or
x	crossing	
y	near (gap)	or or
z	left of	

(b)

Fig. 10. Attribute codes for primitives and relations. (a) Primitive codes. (b) Relation codes.

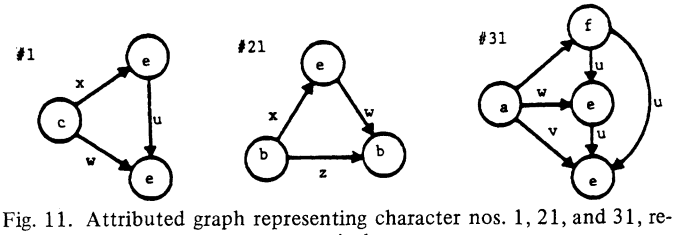


Fig. 11. Attributed graph representing character nos. 1, 21, and 31, respectively.

unknown pattern graphs can be classified by using the proposed distance or by applying the maximum likelihood rule.

Although the distance measure proposed in this paper is conceptually similar to that used in [8], which is based on the numbers of separate transformation operations such as vertex insertion, deletion, and substitution, we have combined the transformation operations into one mapping procedure. Insertion now becomes the mapping of a nonnull element onto a null element, deletion now becomes the mapping of a nonnull element onto a null element, and substitution now involves the

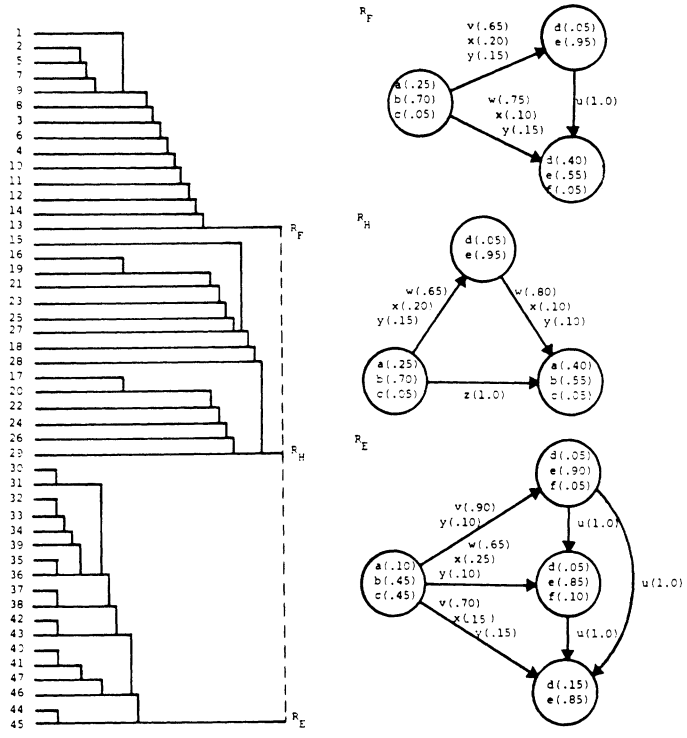


Fig. 12. The hierarchical clustering process. Random graphs representing F's, H's, and E's.

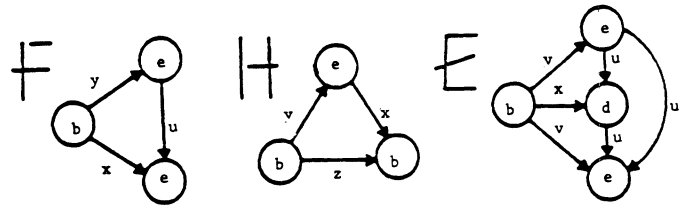


Fig. 13. Three unknown patterns and their attributed graphs.

d	R_F	R_H	R_E
(a)	0.334	0.977	1.411
(b)	0.977	0.334	2.138
(c)	1.366	2.237	0.373

Fig. 14. Distance between pattern graphs and random graphs.

mapping of one nonnull element onto another element with different attribute values.

As mentioned previously, deformed patterns can be considered as members of the class in clustering or recognition. Deformation may include changes in attribute values in both primitives and relations. In some cases, a primitive in a pattern may be split into two, or two separate ones may be merged into one. For example, a character stroke may be broken into two segments after preprocessing, or two "close" segments may be joined into one due to noise. In general, the local deformation (splitting or merging) in patterns would be reflected locally in the determination of optimal morphism. Our random graph approach is generally able to resolve these types of deformation. In cases where splitting and merging occur frequently, graph distance including the transformation of graph element splitting and merging [14], [15] would be necessary.

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To provide good algorithms, various techniques have been developed for attributed (or labeled) graph matching in pattern analysis [7], [16]-[19]. Based on a branch-and-bound tree searching technique, we have developed for graph optimal isomorphism and monomorphism [20] a more efficient algorithm using a heuristic function for a consistent lower bounded cost estimation. This algorithm, together with other related issues, will be reported in a separate paper currently under preparation.

Finally, when dealing with complex structural patterns, it is possible to include n-ary relations by adopting hypergraphs [10] in which a hyperedge may connect more than two vertices. It is also desirable to include the notion of hierarchical graphs [21], [22]. Both hypergraphs and hierarchical graphs can be generalized to random hypergraphs and hierarchical graphs.

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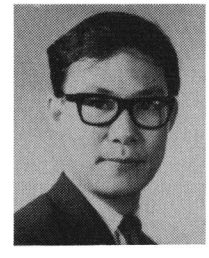

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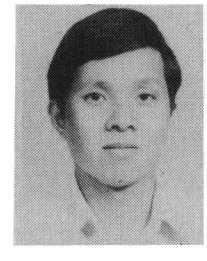

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